Sheikh Mohsin

B.Sc. Computer Science & Business Analytics, FLAME University · Pune, India · Software engineering internships

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Education

FLAME University, Pune · *B.Sc. CS & Business Analytics, full scholarship · CGPA 7.65/10* Aug 2024 – Present

- Modeled and verified concurrent and distributed systems with SAT solvers, temporal-logic model checking, and Alloy/TLA+ specifications
- Implemented and compared classification and clustering methods, using PCA to test how feature selection affected model behavior
- Member, Dot Slash Club; interdisciplinary projects connecting computer science with Philosophy and Public Policy

Experience

IIT Ropar · *Research Intern (Summer Internship Program), SSD Systems* May 2026 – Present

- Compared recency- and frequency-based cache eviction strategies in the SimpleSSD-Standalone 2.0 simulator under Prof. Venkata Kalyan Tavva
- Designed and implemented spatial prefetching that predicts nearby address-translation entries from how data is grouped in flash
- Found and fixed a cache-order bug that caused newly loaded entries to evict one another before use
- Investigated how ransomware write bursts interact with the drive's internal cleanup and identified a race that may destroy recoverable data before detection responds

PartyHub (Remote) · *QA Testing Intern* Jul 2025 – Aug 2025

- Tested frontend and backend features before release, focusing on edge cases that could break real user flows rather than only the happy path
- Wrote reproducible bug reports with exact steps and expected-versus-actual behavior, giving developers a clear path from discovery to triage
- Checked API responses and cross-device behavior to expose integration issues before release

Projects

Cache Performance Benchmark Suite · *C++, `perf_event_open`, RDTSCP, Chart.js* Repository

- Measured hardware clock-cycle costs across 16 data structures with performance counters, isolating machine-level behavior from wall-clock noise
- Compared contiguous (array-of-structs, struct-of-arrays) and pointer-based layouts across access patterns and data sizes
- Automated collection, CSV export, and Chart.js visualization so results are reproducible

IPL Data Mining · *Python, data analysis, visualization* Repository

- Built a repeatable analysis workflow that separates data preparation from interpretation so conclusions can be checked against the underlying records

sheikh-mohsin.vercel.app · *personal website and diary* Repository

- Next.js, TypeScript, and MDX site with a terminal UI, file-based diary, and LaTeX CV pipeline; deployed on Vercel from a template by Jeet Shah

Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS, LaTeX
Systems: Data structures and algorithms, FTL/NAND flash internals, hardware performance profiling (`perf_event_open`, RDTSCP), Linux
Tools: Git/GitHub, Next.js, Vercel, basic AWS, Markdown
Communication: Technical writing, research communication; English, Hindi, Urdu, Kashmiri

Leadership

Karta Initiative India Foundation · *University Leader; Technical and Operations Lead, workshops* Aug 2024 – Present

- Peer representative for 40+ Karta Scholars; ran weekly Sabhas, onboarded new scholars, and escalated concerns to mentors
- Ran technical operations for a five-day winter workshop and coordinated housing, food, events, and schedules for the summer workshop